Huawei Access Node BAS-ACC-37 QoS Congestion Event SOP (ALARM_8171)

Node: BAS-ACC-37 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_8171

Description

The occurrence of ALARM_8171 (Warning) on Huawei Access Node BAS-ACC-37, situated at Basavanagudi POP, is attributed to QoS congestion events precipitated by microbursts, interface tail drops, and QoS queue starvation. This phenomenon is a direct consequence of the node's role as an access point, where the high volume of concurrent traffic demands stringent Quality of Service (QoS) guarantees. The Huawei eNSP (Enhanced Network Service Platform) equipment, upon which this node is based, employs a hierarchical QoS architecture, with the E2 (Edge) layer serving as the entry point for traffic into the network. The E2 layer's QoS functionality relies on the classification and marking of traffic based on 802.1p and DSCP (Differentiated Services Code Point) values, which are then mapped to corresponding QoS queues. However, in scenarios where microbursts or sustained traffic spikes occur, the QoS queues may become congested, leading to interface tail drops and subsequent ALARM_8171 warnings.

Pre-requisite Checks

- Verify that all redundant links are operational and connected.

- Check the node's power and environmental status to ensure that it is functioning within normal parameters.

- Dispatch a remote hands team to the Basavanagudi POP to facilitate access to the node's CLI.

Potential Root Causes

Cause	Probability
Microbursts or sustained traffic spikes causing QoS queue congestion	High
Incorrect QoS configuration or marking mismatches	Medium
Hardware or software issues on the E2 (Edge) layer	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos queue info, Expected Output: Queue information: Queue 1: 100% utilization, Queue 2: 0% utilization, Description: This command will display the current QoS queue utilization, helping to identify which queues are congested.

Command: display interface brief, Expected Output: Interface 1: 100% utilization, Interface 2: 0% utilization, Description: This command will display the current interface utilization, helping to identify which interfaces are experiencing congestion.
```

```
Command: display qos statistics, Expected Output: QoS statistics: 1000 packets dropped due to queue congestion, Description: This command will display QoS statistics, helping to quantify the impact of congestion on the network.
```

Resolution Steps

Step 1: Step: Verify QoS configuration and markings, Command: display qos policy, Description: Verify that the QoS configuration and markings are correct and align with the expected traffic patterns.

Step 2: Step: Adjust QoS queue configuration, Command: qos queue adjust 1 1000, Description: Adjust the QoS queue configuration to increase the buffer size for the congested queue, helping to mitigate congestion.

Step 3: Step: Clear QoS statistics, Command: clear qos statistics, Description: Clear QoS statistics to reset the counters and enable accurate monitoring of congestion.

Post-Check Verification

```
Command: display qos queue info, Expected Output: Queue information: Queue 1: 50% utilization, Queue 2: 0% utilization, Description: Verify that the QoS queue utilization has decreased after adjusting the QoS queue configuration.

Command: display interface brief, Expected Output: Interface 1: 50% utilization, Interface 2: 0% utilization, Description: Verify that the interface utilization has decreased after adjusting the QoS queue configuration.
```

Roll-back Procedures

- Step: Restore original QoS configuration, Command: qos queue restore, Description: Restore the original QoS configuration to ensure that the network returns to its previous state.

- Step: Clear QoS statistics, Command: clear qos statistics, Description: Clear QoS statistics to reset the counters and enable accurate monitoring of congestion.